

# Mean, Median, and Baselines

Project: Responsible Plant Classifier

## 1. Your mission

### Your learning target

I can fit mean and median constant baselines on training targets.

I can compare constant predictions on separate records.

I can keep fitted constants independent of evaluation answers.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

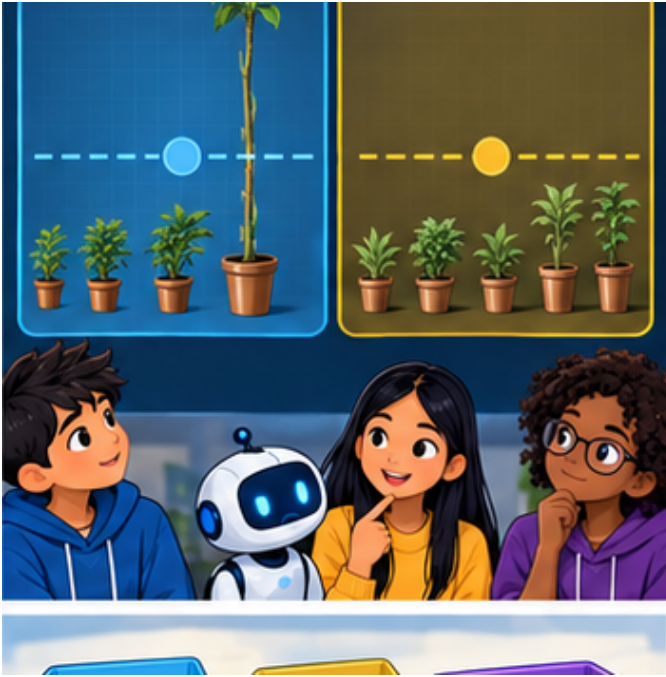
Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Mean and median are transparent regression baselines; outliers affect them differently.



LEARNING TARGET For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

Builder name and experiment date:

### Readiness for the connected explanation

Why can signed residuals hide error size?

### This week's end goal

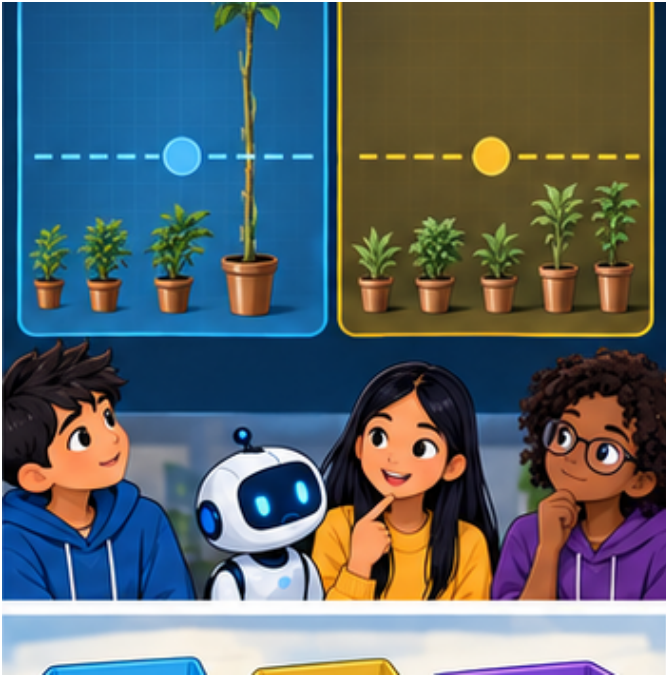
By the end, I can explain mean, median, and baselines using a traced example and a stated limitation.

## 2. Notice the evidence

PICTURE ENTRY

**Notice** → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



**MYSTERY** The team must use mean, median, and baselines to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

### A closer explanation

Before a complex regression model, fit a simple constant predictor. A mean baseline uses the training average. A median baseline uses the middle training target. They ignore features but provide a comparison that is easy to reproduce. Evaluation targets must not determine the constants.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines constant prediction baselines.

## 3. See how it works

ML MECHANISM

**Input → Model → Evidence**

Trace the information, fitted behavior, and evaluation proof separately.



**IN**

## **INPUT**

Choose and justify a baseline under an extreme value.



**ML**

## **MODEL**

Mean and median are transparent regression baselines; outliers affect them differently.



OUT

## EVIDENCE

Choose and justify a baseline under an extreme value.

QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA Mean and median are transparent regression baselines; outliers affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. Mean =  $100 \div 5 = 20$  cm.

Draw a fresh data → model → prediction → evaluation trace:

### A closer explanation

Training targets [2, 4, 6, 12] have mean  $24/4=6$  and median  $(4+6)/2=5$ . On separate targets [4, 8], the mean baseline predicts [6, 6] with absolute errors [2, 2], averaging 2. The median baseline predicts [5, 5] with errors [1, 3], also averaging 2.

```
from statistics import mean, median

training = [2, 4, 6, 12]
validation = [4, 8]

for name, value in [("mean", mean(training)),
                    ("median", median(training))]:
    mae = mean(abs(y - value) for y in validation)
    print(name, value, mae)

# Both validation MAEs are 2; constants differ.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses

different evidence.

## Connect the picture and the explanation

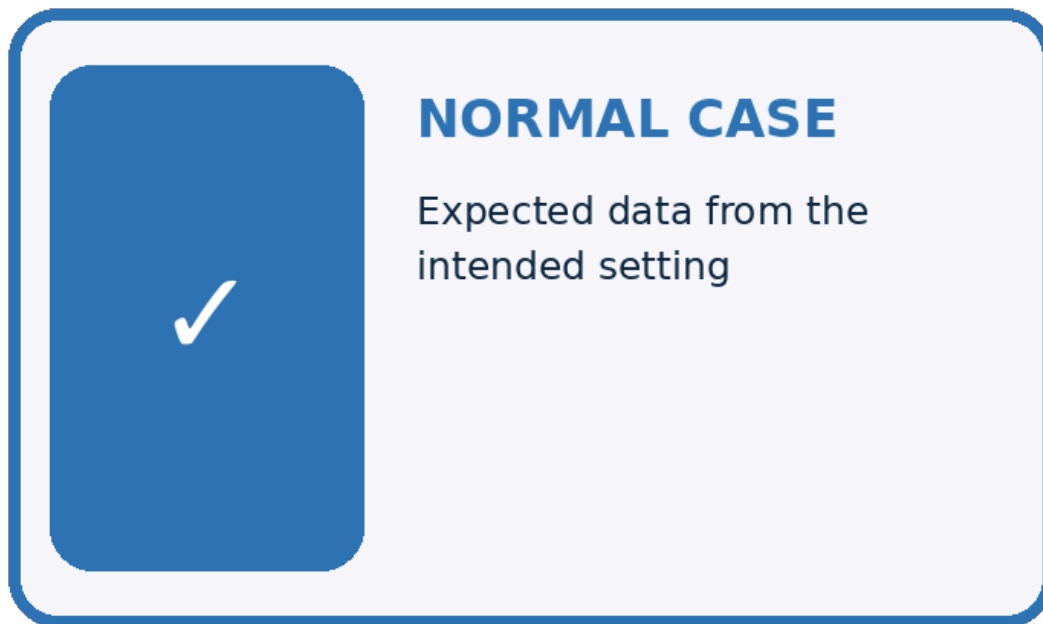
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### EVALUATION DESIGN

#### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





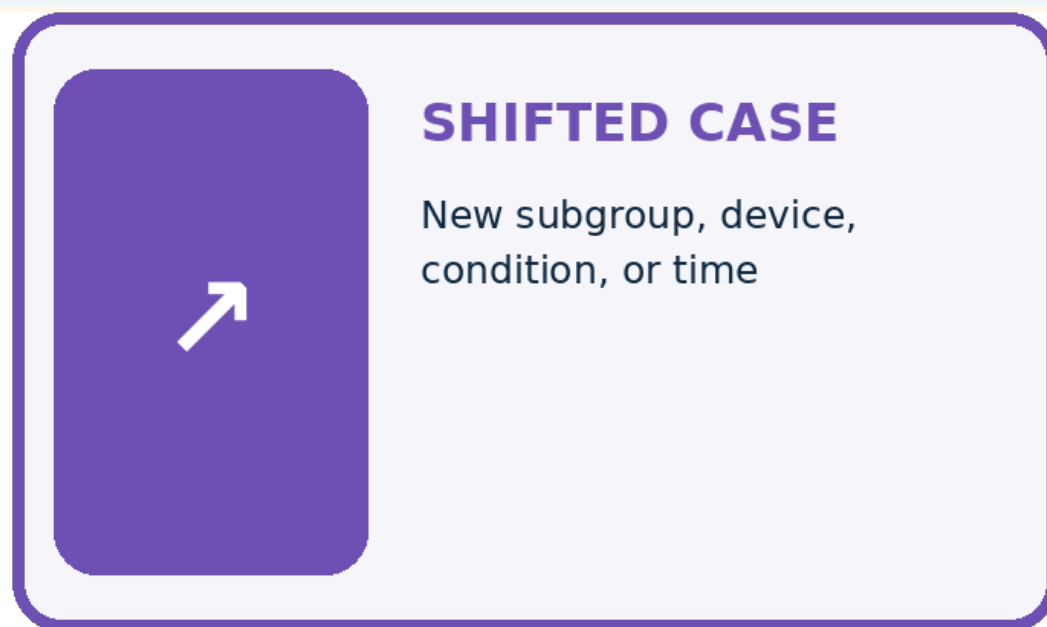
## **BOUNDARY CASE**

Value near a threshold or class boundary



## **MISSING CASE**

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

### Change one thing in the pictured case

Change: Replace training 20 with 8 while keeping evaluation unchanged.

Prediction: Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm.

Reason: Mean is now  $20/4=5$ ; median stays 5. The earlier training outlier affected the mean.

## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

## MEAN

sum divided by count

2

## MEDIAN

middle ordered value

3

## OUTLIER

value far from most observations

4

## LOCATION BASELINE

single typical value predicted for all cases

**Term · Precise meaning · My ML example**

**Term:** mean

**Precise meaning:** sum divided by count

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** median

**Precise meaning:** middle ordered value

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** outlier

**Precise meaning:** value far from most observations

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** location baseline

**Precise meaning:** single typical value predicted for all cases

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

MISCONCEPTION TO REPAIR The mean is always the fairest and most scientific summary.

Repair the claim using a definition, calculation, or test record:

### Words used in the closer explanation

#### Word · Meaning

**Word:** Constant baseline

**Meaning:** One fixed prediction for every query

\_\_\_\_\_

**Word:** Training mean

**Meaning:** Sum of training targets divided by their count

\_\_\_\_\_

**Word:** Training median

**Meaning:** Middle value of sorted training targets

\_\_\_\_\_

**Word:** Absolute error

**Meaning:** Magnitude of observed minus predicted

\_\_\_\_\_

## 6. Follow a worked example

## WORKED EVIDENCE

### Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

#### WEEK 20 • WORKED EVIDENCE

- 1 Heights are 10, 11, 12, 13, and 54 cm.
- 2 Mean =  $100 \div 5 = 20$  cm.
- 3 Median is 12 cm, the middle ordered value.
- 4 For typical-height absolute error, median is more robust; still report why 54 is present.

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

STEP 2 Mean =  $100 \div 5 = 20$  cm.

STEP 3 Median is 12 cm, the middle ordered value.

STEP 4 For typical-height absolute error, median is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

### A closer explanation

### Constant prediction baselines at a glance

#### Part · Evidence or action · Interpretation

**Part:** Training targets

**Evidence or action:** Fit mean 6

**Interpretation:** Freeze constant

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**Part:** Training targets

**Evidence or action:** Fit median 5

**Interpretation:** Freeze constant

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**Part:** Evaluation targets

**Evidence or action:** Compare fixed predictions

**Interpretation:** Both MAE 2 here

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Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

**Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.**

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## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Mean, Median, and Baselines

Fit two constant height baselines and compare them on held-out examples.

## The facts we will use

Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from training.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Mean, Median, and Baselines

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Fit mean

Training mean= $(2+4+6+20)/4=8$  cm.



#### 2. Fit median

Training median= $(4+6)/2=5$  cm.



#### 3. Evaluate constants

Mean MAE= $(4+2)/2=3$  cm; median MAE= $(1+5)/2=3$  cm.

### CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Fit mean

**What happens and why:** Training mean= $(2+4+6+20)/4=8$  cm.

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**Step:** 2. Fit median

**What happens and why:** Training median= $(4+6)/2=5$  cm.

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### Step: 3. Evaluate constants

**What happens and why:** Mean  $MAE=(4+2)/2=3$  cm; median  $MAE=(1+5)/2=3$  cm.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. Mean  $MAE=(4+2)/2=3$  cm; median  $MAE=(1+5)/2=3$  cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

Mean and median have different loss properties, but their held-out performance depends on the particular evaluation cases.

Read the labelled picture: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Replace training 20 with 8 while keeping evaluation unchanged. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

ERROR DETECTIVE

**Compare → Diagnose → Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



### COMPARE • DIAGNOSE • REPAIR

Tempting claim: The mean is always the fairest and most scientific summary. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** passes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** defined behavior

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing case

**Expected:** safe fallback

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** shifted case

**Expected:** measured limit

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

CLAIM TO CHALLENGE The mean is always the fairest and most scientific summary.

Diagnose one cause and name the cleanest next experiment:

## A closer explanation

For an even number of values, this course averages the two middle values to report one median. Keep a frozen baseline across comparisons and show its evaluation score. A model that fails to beat it may still reveal a data or metric issue worth investigating.

**Why not predict the evaluation mean as the baseline?**

\_\_\_\_\_  
\_\_\_\_\_

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## 8. Practise together

ML STUDIO

### Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



#### **FREEZE THE EXPERIMENT**

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

### Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

### A closer explanation

Replacing training 12 by 40 gives mean 13 and median 5. That change is a data change to record and investigate, not an instruction to delete an inconvenient observation. Decide whether the extreme value is valid using the measurement policy and provenance.

### Practise with less help: Score separately

Use workbook W2 for the connected example “Constant prediction baselines”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Constant prediction baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EVIDENCE LAB

**Predict → Run → Calculate → Interpret**

Keep the test conditions fixed, record every result, and change only one planned factor.

# LAB

## PREDICT → RUN → CALCULATE → INTERPRET

Baseline Tournament: compute mean and median, predict every case, compare MAE, and explain the outlier's effect.

1. Write the hypothesis and expected evidence before running.
2. Baseline Tournament: compute mean and median, predict every case, compare MAE, and explain the outlier's effect.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Case · Prediction · Actual · Calculation / evidence · Meaning / next test

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Shifted

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

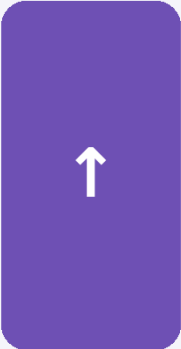
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

### From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



**ML BUILDER MASTERY**  
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

**DEFINE** Use all four terms precisely: mean, median, outlier, location baseline.

**CALCULATE** Reproduce the worked evidence with units and denominator.

**TRACE** Explain the data, code or decision path.

**DIAGNOSE** The mean is always the fairest and most scientific summary.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

## 11. Connect your project

CUMULATIVE PROJECT

### Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

THIS WEEK'S CHECKPOINT Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

## Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 12. Show what you know

INDEPENDENT PROOF

**Explain • Calculate • Transfer • Defend**

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



### INDEPENDENT ML BUILDER PROOF

Choose and justify a baseline under an extreme value.

REQUIRED PROOF Choose and justify a baseline under an extreme value.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Absolute Error and MAE.

## 13. Study a complete case

## Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

Training heights are 2,4,12 cm. Compare constant mean and median baselines.

### The question

Compute both and training absolute errors.

### Worked explanation

Mean=6, median=4. Mean errors 4,2,6 total12; median errors 2,0,8 total10. Median minimises absolute error here.

### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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### Trace the supplied case visually

#### Mean and median baselines use the same observations

**Evidence or stage · Value or result**

**Evidence or stage:** Observation A

**Value or result:** 2

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**Evidence or stage:** Observation B

**Value or result:** 4

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**Evidence or stage:** Observation C

**Value or result:** 12

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**Evidence or stage:** Mean baseline

**Value or result:** 6

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**Evidence or stage:** Median baseline

**Value or result:** 4

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Training heights are 2,4,12 cm. Compare constant mean and median baselines.

Compute both and training absolute errors.

Mean=6, median=4. Mean errors 4,2,6 total12; median errors 2,0,8 total10. Median minimises absolute error here.

## 14. Try a new case independently

### Independent case: Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For training heights 2, 3 and 10 cm, calculate mean and median. Which constant is less influenced by the high value?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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## Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

**Transfer practice from the original programme**

For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

**Explain the mechanism, evidence and one limit.**

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**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

**15. Your lesson route**

Start with this question: Fit two constant height baselines and compare them on held-out examples.

**By the end, I can · How I will show it**

**By the end, I can:** Fit mean and median constant baselines on training targets  
**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Compare constant predictions on separate records  
**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Keep fitted constants independent of evaluation answers

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Fit mean and median constant baselines on training targets

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit two constant height baselines and compare them on held-out examples.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean  $MAE = (4+2)/2 = 3$  cm; median  $MAE = (1+5)/2 = 3$  cm.

1. Fit mean
2. Fit median
3. Evaluate constants

### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Evaluate constants Mean  $MAE = (4+2)/2 = 3$  cm; median  $MAE = (1+5)/2 = 3$  cm. Read the picture in order. Mean  $MAE = (4+2)/2 = 3$  cm; median  $MAE = (1+5)/2 = 3$  cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 2: explain the deciding evidence

Changed-case practice: Replace training 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?

1. Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm. Mean is now  $20/4 = 5$ ; median stays 5. The earlier training outlier affected the mean.

### Repair a missing building block

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric prediction from classification. Short practice: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

## **Predict, run and debug**

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

## **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.